

We were told by a TA that we don't need to provide a write-up for the experiments and just do the take-home exercise.

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Exercise:

The first thing that we did was take the code from experiment 5 and use it as a base to build from to complete the take-home exercise. An important note to mention is that a flaw in our experiment 5 code was found in the counter logic. Before the counter would stay at 00 until the count up or count down button was pressed. To fix that we set the "up" to 1 instead of 0 that way the counter would automatically go into the if statements that the counter was in up mode. Now talking about the exercise after that logic was fixed we set up the buttons to perform as the lab document described us to. Button 0 was to stop and start the counter, button 1 was to count up, button 2 was to count down, and button 3 was to set the counter to 33 only when the counter was stopped. We programmed each button to set a flag variable to true when the button was pressed and used that in the counter's logic to know which action the counter should perform. A keynote is that the counter's boundaries were set to base 6 instead of base 10. To do that we changed the edge cases to deal with it. Let's talk about the count-up mode. The edge cases we need to deal with are when both digits are 5 we set both of the digits to 00, dealing with the case when the counter reaches the maximum value and it needs to roll back to 00 to start the cycle again. The next edge case we needed to deal with was when the right digit counted up to 5 it needed to be set back to 0 and add one to the left digit. If neither of these conditions were met it just continued to count up. Moving onto the countdown logic it is very similar to the count-up logic. The first case was when both digits were 0 it would roll to the maximum number 55 and if the right digit was 0 we would set it back to 5 and subtract one from the left digit if none of the conditions were met it would continue to count down. The last counter logic was when the counter has to be stopped and button 3 is pressed which will set the counter to display 33 and that was done with a simple if condition. For the LED lights all of that was done using logic gates truth table and correlating it to each independent LED with if statements to separate each one depending on what it was being asked for. Initial status of all LED lights are off (0'd1), and the if statements depict the certain logic needed to turn them on.